

AI License Plate Recognition

Moving Vehicle Input -> Plate Recognition & Display

Regular Course / AI Solution for Traffic Safety

Project Goal

- Input: Moving vehicle files (video / images)
- Output: Recognize and display license plate numbers
- Path: C:\tool\W\yolo26-main\Wimage (e.g. img1.png, img2.png, img3.png)

Pipeline

1. Roboflow: Upload & label vehicle/plate images
2. Dataset: Generate train/val/test from labeled data
3. YOLO: Train model -> best.pt
4. Python: Load best.pt, run detection on input files
5. ROI / CROP: Extract plate region from frame
6. CLAHE: Contrast enhancement for OCR
7. Homography: Perspective correction (exam_realtime_plate.py)
8. OCR: PaddleOCR + EasyOCR -> plate text
9. Display: Bbox + recognized plate number on screen

Evaluation Criteria (100 pts)

- Self-assessment report: 40 pts
- Labeling work: 5 pts (Roboflow -> dataset)
- Dataset creation: 5 pts (data.yaml, train/val split)
- Vehicle/plate detect on input file: 10 pts (YOLO bbox)
- ROI, CROP, CLAHE, Homography: 20 pts (plate_engine_pro.py, exam_realtime_plate.py)
- License plate recognition display: 20 pts (plate text + confidence on screen)

Code tags: [평가기준 10점], [평가기준 20점] in plate_engine_pro.py, plate_server.py

Result Example

Recognized plates (confidence):

- 01나8060 (71%)
- 경기91바6286 (79%)
- 서울바9203 (72%)

Output: Red/green bbox on plate + plate text above.

See: plate_server.py (localhost:5000), plate_engine_pro.process_frame()

Summary

- YOLO (best.pt) detects plate region -> ROI/CROP
- CLAHE + optional Homography improve readability
- PaddleOCR + EasyOCR ensemble -> Korean plate text
- Validation & voting -> final plate string
- Display: plate number + confidence on video/image.

Files: plate_engine_pro.py, plate_server.py, exam_realtime_plate.py

Mapping: 평가기준_기능매핑.md